

STATISTICS

PART 3: Quartiles, Boxplots, Relative Measures of Distributions, Skewness Measures

Birol Aslanyürek

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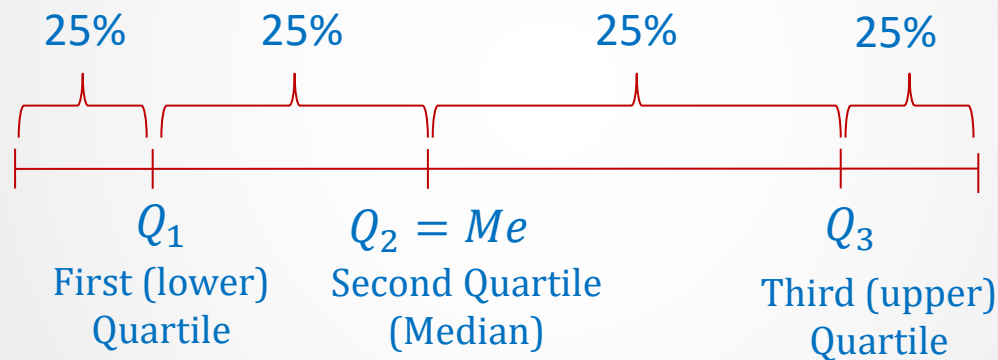
Quartiles

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Quartiles

Quartiles consist of three measures that divide the ordered data into four equal parts.



Quartiles can be computed by

$$Q_1 = X_{\frac{n+1}{4}}$$

$$Q_2 = X_{\frac{2(n+1)}{4}} = X_{\frac{n+1}{2}}$$

$$Q_3 = X_{\frac{3(n+1)}{4}}$$

where n is the number of data and X_i is the i^{th} value in the ordered data set.

Quartiles

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Example 1: The number of pulses for 11 athletes are given as:

$$X = \{101, 103, 108, 112, 96, 105, 107, 97, 102, 104, 106\}$$

Find the quartiles of the data.

Solution:

The ordered data:

$$X = \{96, 97, 101, 102, 103, 104, 105, 106, 107, 108, 112\}$$

$$Q_1 = X_{\frac{n+1}{4}} = X_{\frac{11+1}{4}} = X_3 = 101$$

$$Me = Q_2 = X_{\frac{n+1}{2}} = X_{\frac{11+1}{2}} = X_6 = 104$$

$$Q_3 = X_{\frac{3(n+1)}{4}} = X_{\frac{3(11+1)}{4}} = X_9 = 107$$

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Example 2: Let us add 98 to the data set given in the previous example:

$$X = \{101, 103, 108, 112, 96, 105, 107, 97, 102, 104, 106, 98\}$$

Find the quartiles of the data.

Solution:

The ordered data:

$$X = \{96, 97, 98, 101, 102, 103, 104, 105, 106, 107, 108, 112\}$$

For the non-integer index, we can use linear interpolation.

$$Q_1 = X_{\frac{n+1}{4}} = X_{\frac{12+1}{4}} = X_{3.25} = X_3 + (X_4 - X_3)0.25 = 98 + (101 - 98)0.25 = 98.75$$

$$Q_2 = X_{\frac{n+1}{2}} = X_{\frac{12+1}{2}} = X_{6.5} = \frac{X_6 + X_7}{2} = \frac{103 + 104}{2} = 103.5$$

$$Q_3 = X_{\frac{3(n+1)}{4}} = X_{\frac{3(12+1)}{4}} = X_{9.75} = X_9 + (X_{10} - X_9)0.75 = 106 + (107 - 106)0.75 = 106.75$$

Quartiles

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Example 3: Find the quartiles for the puzzle example.

Answer: $Q_1 = 10$, $Q_2 = 13.5$, $Q_3 = 16.25$

When the CFD is given, the quartiles can be obtained as follows:

$$Q_i = L_i + \frac{\frac{i n}{4} - (\sum f)_i}{f_{Q_i}} c_i, \quad i = 1, 2, 3$$

L_i : The lower bound of the class containing Q_i

n : The number of data

$(\sum f)_i$: The sum of the frequencies of the classes that are smaller than the class containing Q_i

f_{Q_i} : The frequency of the class containing Q_i

c_i : The interval of the class containing Q_i

Quartiles

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$$\text{Interquartile Range} = Q_3 - Q_1$$

Example 4: The CFD of 70 students' weight in a class is given below. Find the first and third quartiles. Then determine the interquartile range.

Weight (C_i)	Frequency (f_i)	Cumulative Frequency (F_i)
44-51	8	
51-58	11	
58-65	17	
65-72	15	
72-79	10	
79-86	6	
86-93	3	

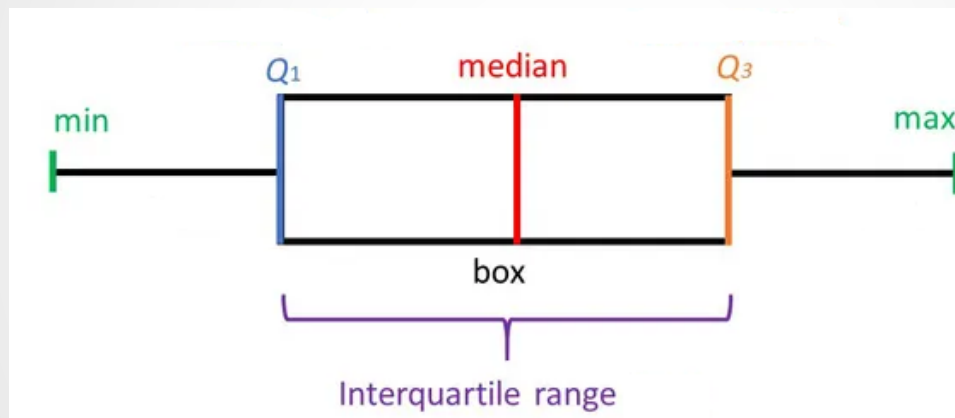
Boxplots

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Boxplots

Boxplots illustrate the range, outliers, quartiles and symmetry of data.



Interpretation regarding Boxplots:

1. Symmetry of data between interquartile range (50% of data around the central point):

$|Me - Q_3| \gg |Me - Q_1| \Rightarrow$ The data is skewed to right

$|Me - Q_3| \ll |Me - Q_1| \Rightarrow$ The data is skewed to left

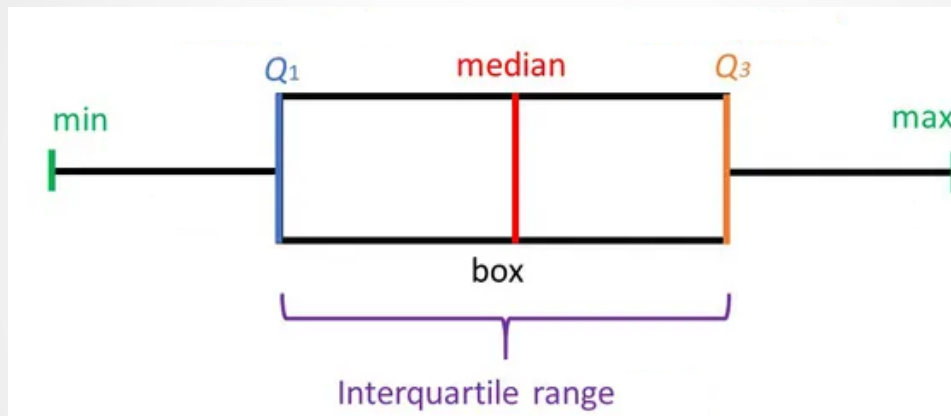
$|Me - Q_3| \cong |Me - Q_1| \Rightarrow$ The data is symmetric

Boxplots

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Boxplots



Interpretation regarding Boxplots:

2. Symmetry of all data (100% of data):

- $|Me - Q_3| \gg |Me - Q_1|$ and $|\max - Q_3| \gg |Q_1 - \min|$ $\Rightarrow$ The data is skewed to the right
- $|Me - Q_3| \ll |Me - Q_1|$ and $|\max - Q_3| \ll |Q_1 - \min|$ $\Rightarrow$ The data is skewed to the left
- $|Me - Q_3| \cong |Me - Q_1|$ and $|\max - Q_3| \cong |Q_1 - \min|$ $\Rightarrow$ The data is symmetric

Boxplots

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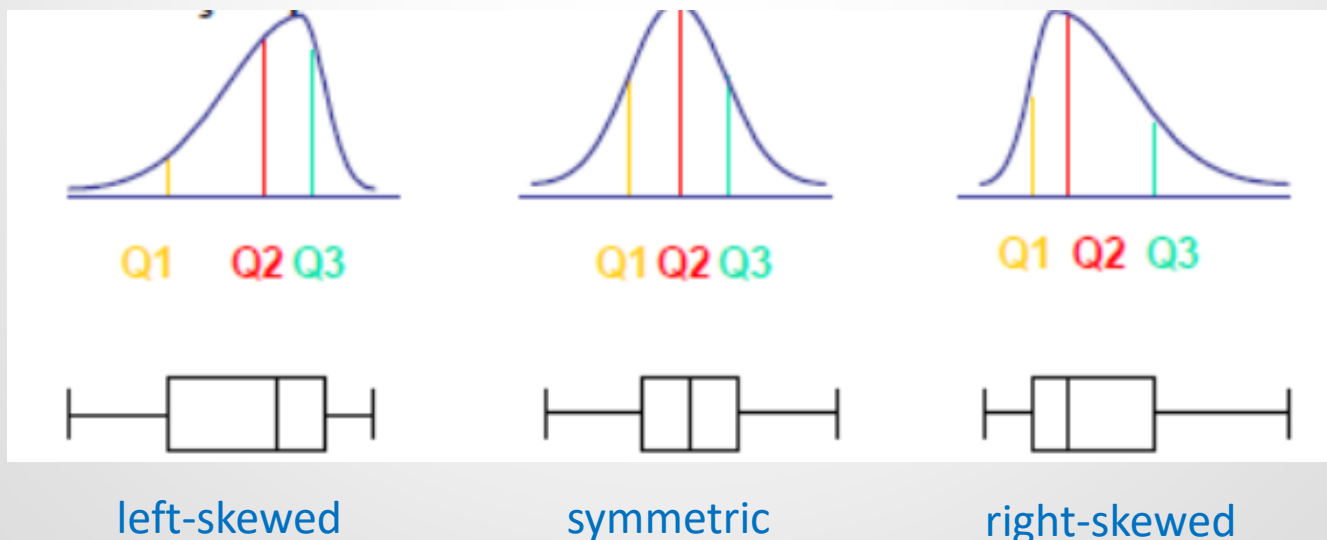
Boxplots

2. Symmetry of all data (100% of data):

$|Me - Q_3| \gg |Me - Q_1|$ and $|\max - Q_3| \gg |Q_1 - \min|$ $\Rightarrow$ The data is skewed to the right

$|Me - Q_3| \ll |Me - Q_1|$ and $|\max - Q_3| \ll |Q_1 - \min|$ $\Rightarrow$ The data is skewed to the left

$|Me - Q_3| \cong |Me - Q_1|$ and $|\max - Q_3| \cong |Q_1 - \min|$ $\Rightarrow$ The data is symmetric

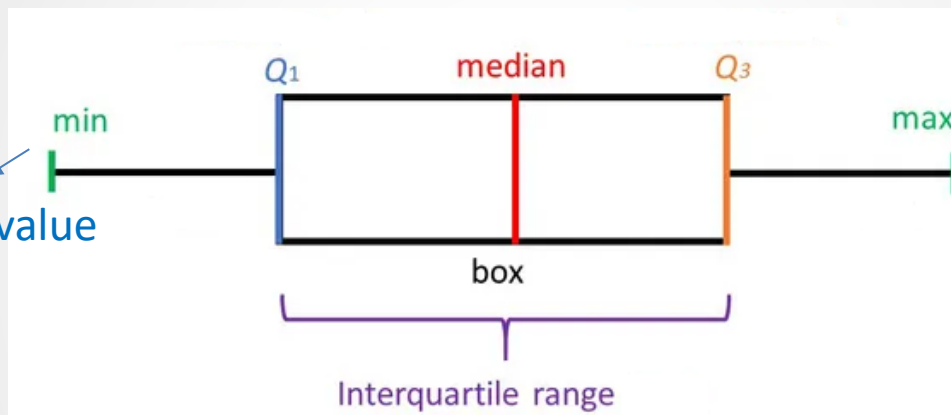


Boxplots

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Boxplots



The updated min value

The updated max value

Interpretation regarding Boxplots:

3. Outliers (Extreme values):

$|\max - Q_3| > 1.5|Q_3 - Q_1|$ $\Rightarrow$ The data contain outliers on the right side.

$|Q_1 - \min| > 1.5|Q_3 - Q_1|$ $\Rightarrow$ The data contain outliers on the left side.

If the data contain outliers, the horizontal line ending with the minimum (or maximum) value is shortened according to a new minimum (or maximum) value by excluding the outliers.

Boxplots

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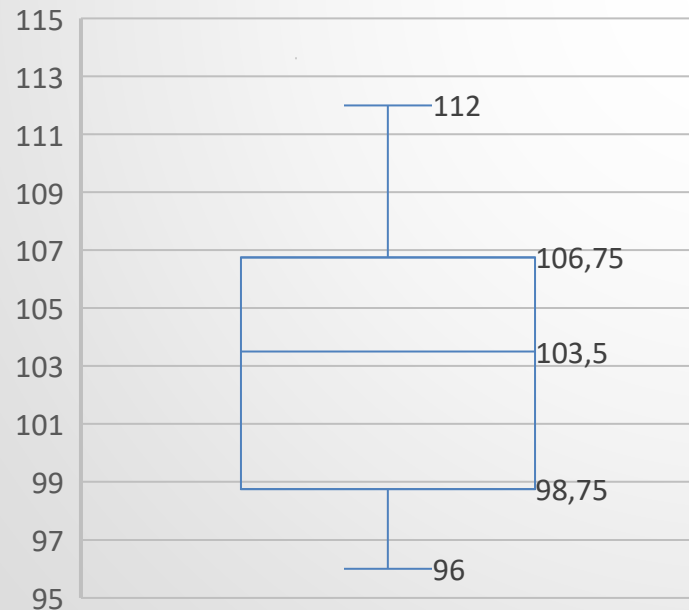
Example 5: Illustrate the boxplot for the data given in example 2. Interpret the result.

$$X = \{96, 97, 98, 101, 102, 103, 104, 105, 106, 107, 108, 112\}$$

$$Q_1 = 98.75$$

$$Q_2 = 103.5$$

$$Q_3 = 106.75$$



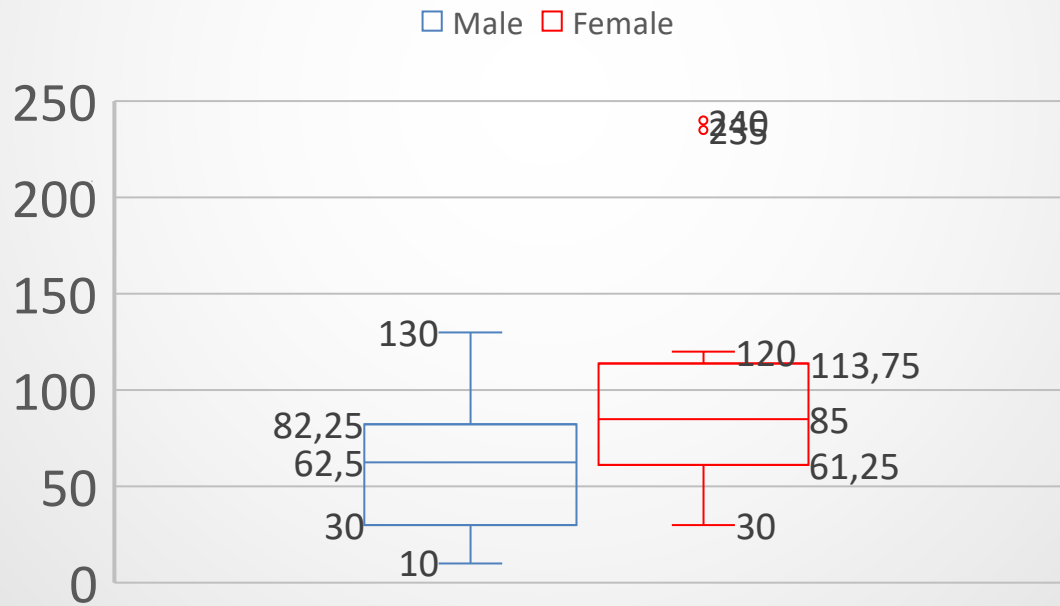
Boxplots

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Example 6: The duration of watching television (in mins.) is given below for 20 males and 20 females. Illustrate the boxplot and interpret the results.

Male	Female
30	100
45	120
60	80
20	95
10	110
30	235
65	36
68	75
90	90
18	98
35	120
25	30
83	48
65	60
70	65
130	80
90	115
90	240
46	50
80	70



Relative Measures of Distributions

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Relative Measures of Distributions

It may not be right to compare measures of two distributions when their units or observation magnitudes are different (for example, weight vs height, height vs length of shoulders). In this case, we can use relative measures of distributions.

$$\text{Relative Distribution Measure} = \frac{\text{Absolute Distribution Measure}}{\text{Mean}} \times 100$$

If the absolute distribution measure is the standard deviation, and the mean is the arithmetic mean, then this measure is called the **distribution coefficient**, namely

$$D.C. = \frac{\sigma}{\mu} \times 100$$

If $(D.C.)_1 > (D.C.)_2$ for two different data set, the dispersion (distribution) is more in the first data set.

Relative Measures of Distributions

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Example 7: The mean and standard deviation of two shares are given as:

$$\mu_1 = 7 \text{ millions TL}$$

$$\mu_2 = 2 \text{ millions TL}$$

$$\sigma_1 = 0.6 \text{ millions TL}$$

$$\sigma_2 = 0.5 \text{ millions TL}$$

- Compare the shares according to their absolute and relative distribution.
- Which share is riskier?

Relative Measures of Distributions

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Consider the data set $X = \{x_1, x_2, \dots, x_n\}$. **Standard variable** is defined as

$$z_i = \frac{x_i - \mu}{\sigma}$$

where μ and σ are the mean and standard deviation of X , respectively.

$z_i = 0$  x_i is the mean of X

$z_i > 0$  x_i is larger than the mean of X

$z_i < 0$  x_i is smaller than the mean of X

We can use a standard variable to compare two values which are selected in two different distributions.

Relative Measures of Distributions

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Example 8: A student got 72 points in Statistics and 60 points in Mathematics. The mean and standard deviation are $\mu = 43$ and $\sigma = 8$ for Statistics, $\mu = 28$ and $\sigma = 5$ for Mathematics. For which courses, the student is more successful?

Example 9: The average price of the 1996 model of a particular model car is 4 billion, while the average price of the 1994 model of the same brand is 2.4 billion. The standard deviation of prices is 520 million and 240 million, respectively. Is it more advantageous to buy a 1996 car with a price of 3.7 billion or a 1994 car with a price of 2.2 billion? (other features of cars will not be considered)

Answer: $z_{1996} = -0.58$, $z_{1994} = -0.83$,

The price of the car with model 1994 is more advantageous.

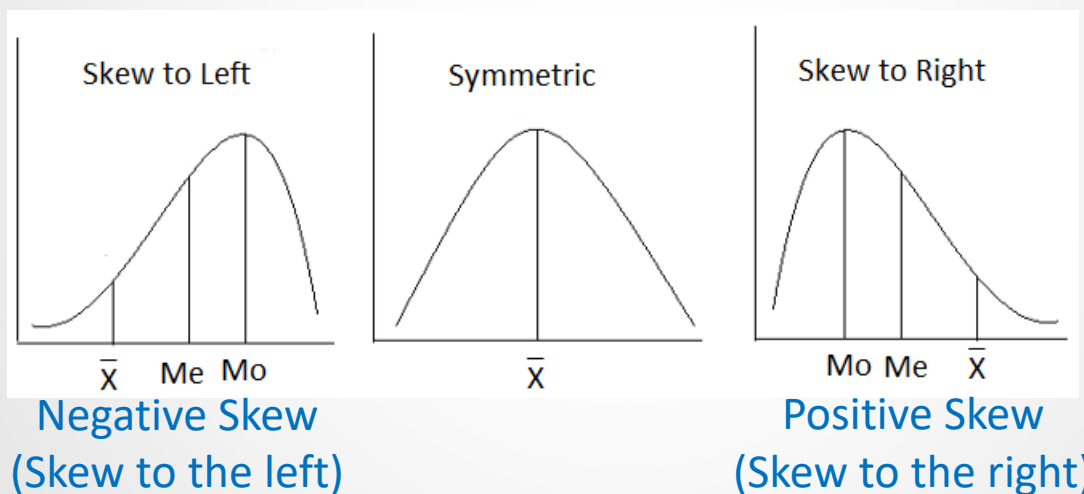
Skewness Measures

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Skewness Measures

Skewness is a measure of symmetry or lack of symmetry.



If we know skewness, we can determine $\mu > Me$ or $\mu < Me$. On the other hand, if we know the order of μ and Me , we can not determine skewness exactly.

Skewness Measures

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Below, some skewness measures are introduced although they are not reliable.

1. Pearson's Skewness Coefficients

$$P_{SC} = \frac{\mu - Mo}{\sigma}$$

Pearson's first skewness coefficient

$$P_{SC} = \frac{3(\mu - Me)}{\sigma}$$

Pearson's second skewness coefficient

The values of Pearson's first skewness coefficient **generally** changes between $-3 < P_{SC} < 3$.

$P_{SC} \approx 0$  Distribution is symmetric.

$P_{SC} > 0$  Distribution is skewed to the right. (data is positively skewed)

$P_{SC} < 0$  Distribution is skewed to the left. (data is negatively skewed)

Example 10: For the data given in example 2, determine and interpret Pearson's second skewness coefficient.

$$X = \{96, 97, 98, 101, 102, 103, 104, 105, 106, 107, 108, 112\}$$

Skewness Measures

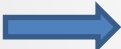
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2. Bowley's Skewness Coefficient

$$B_{SC} = \frac{Q_3 + Q_1 - 2Me}{Q_3 - Q_1}, \quad -1 < B_{SC} < 1$$

$B_{SC} \approx 0$  Distribution is symmetric.

$B_{SC} > 0$  Distribution is skewed to the right. (data is positively skewed)

$B_{SC} < 0$  Distribution is skewed to the left. (data is negatively skewed)

Example 11: For the data given in example 2, determine and interpret Bowley's skewness coefficient.

$$X = \{96, 97, 98, 101, 102, 103, 104, 105, 106, 107, 108, 112\}$$

Skewness Measures

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Kurtosis (Excess Kurtosis) of a Distribution

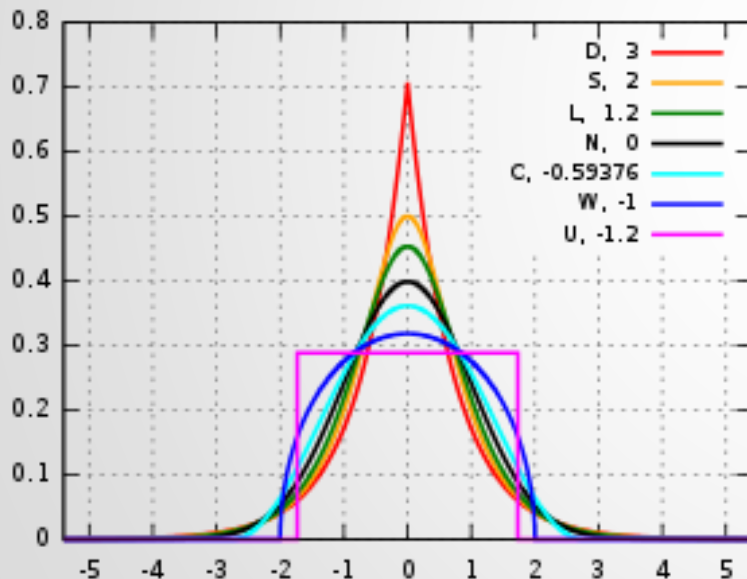
Kurtosis is related to the number of outliers in a data set. It is defined as:

The fourth moment about the mean

$$K = \frac{M_4}{\sigma^4} - 3$$

$$M_k = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^k \quad (\text{for exact case})$$

$$M_k = \frac{1}{\sum_{i=1}^m f_i} \sum_{i=1}^m f_i (x_i - \bar{x})^k \quad (\text{for classified case})$$



Probability density functions for selected distributions with mean 0, variance 1 and different excess kurtosis.

The kurtosis of a normal distribution is 0.

Example

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Example 12: The salaries of 100 workers are given as shown in the table.

- a) Compute the mean and standard deviation.
- b) Compute the mode and median. Interpret results.
- c) Compute the number of data in the interval $\bar{x} \pm \sigma$ and $\bar{x} \pm 2\sigma$.

Salary (C_i)	Number of Workers (f_i)
120-125	10
125-130	20
130-135	38
135-140	25
140-145	7